

Analytic gradients for the exact-arc contact law

The derivative of `roundedContact`'s two functionals with respect to the body arrays, in closed form. Both were previously taken by complex step, one whole evaluation per degree of freedom; this is what replaces that. [Blue marks remarks rather than derivation.](#)

1 The two derivative arguments

The boundary of a rounded body is n circular arcs alternating with n straight segments, meeting tangentially, so it is C^1 with no vertices at all. Write $\partial A \cap B$ as a disjoint union of sub-stretches on each of which the nearest feature of B is a single segment or a single arc; that partition is what `substretches` returns.

Energy. $E = \int_{\partial A \cap B} \frac{k}{3} d_B^3 d\ell$. Differentiating moves the breakpoints, but every Leibniz term vanishes: at a span endpoint $d_B = 0$, and at an interior feature switch d_B is continuous, so the terms from the two adjacent sub-stretches cancel. The derivative is therefore the derivative at *frozen* combinatorics, and the argmin and root solving that produced the partition never have to be differentiated.

Area. Green's integrand does *not* vanish at a crossing, so the same argument fails — freezing the partition is measured 40–60% wrong. Use the shape derivative instead,

$$\frac{\partial}{\partial p} |A \cap B| = \int_{\partial A \cap B} \vec{v}_A \cdot \vec{n}_A ds + \int_{\partial B \cap A} \vec{v}_B \cdot \vec{n}_B ds, \quad \vec{v} = \left. \frac{\partial \vec{x}}{\partial p} \right|_{\text{fixed parameter}}, \quad (1)$$

which has no boundary terms at all; the moving crossings enter only through the domains, and a point is of measure zero. So the spans may be frozen here too, but for a different reason and through a different formula.

2 Body parameters

Differentiate with respect to the arrays a body is stored in rather than the backbone: centers $\vec{z}_k$, radii ρ_k , sweeps ψ_k , and the kiss points $\vec{a}_k^+ = \text{tail}[k]$, $\vec{a}_{k+1}^- = \text{head}[k]$. They are treated as *independent* here even though they are not; imposing the relations between them is the separate per-body pass of §6. That is what makes each case below a plain chain rule with no geometry hidden inside it.

An arc is parametrized by rotation of $\vec{w}_0 = \text{head}[k-1] - \vec{z}_k$ rather than by absolute angle,

$$\vec{x}(t) = \vec{z}_k + \cos(\psi_k t) \vec{w}_0 + \sin(\psi_k t) J \vec{w}_0, \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad t \in [0, 1], \quad (2)$$

so no `arctan2` ever appears. A segment is $\vec{x}(t) = \vec{a}_k^+ + t \vec{e}_k$ with $\vec{e}_k = \text{head}[k] - \text{tail}[k]$.

3 Energy: the four integrals

Let $K\vec{e} = (e_y, -e_x)$, so the inward unit normal of a segment of B is $\vec{n} = K\vec{e}/|\vec{e}|$. Signed distances, positive inside B .

(1) Segment along, segment to. With $\alpha = \vec{n} \cdot (\text{tail}_B - \vec{a}^+)$ and $m = \vec{n} \cdot \vec{e}$, the contribution is $|\vec{e}| P$ with $P = \int_{t_0}^{t_1} (\alpha - mt)^3 dt$. Expanded (*not* as $-[(\alpha - mt)^4]/4m$, which divides by zero whenever the two segments are parallel):

$$P = \alpha^3 \Delta_1 - \frac{3}{2} \alpha^2 m \Delta_2 + \alpha m^2 \Delta_3 - \frac{1}{4} m^3 \Delta_4, \quad \Delta_j = t_1^j - t_0^j, \quad (3)$$

$$\frac{\partial P}{\partial \alpha} = 3\alpha^2 \Delta_1 - 3\alpha m \Delta_2 + m^2 \Delta_3, \quad \frac{\partial P}{\partial m} = -\frac{3}{2} \alpha^2 \Delta_2 + 2\alpha m \Delta_3 - \frac{3}{4} m^2 \Delta_4. \quad (4)$$

(2) Segment along, arc to. `pairEnergy` writes this as $\sqrt{q_a t^2 + q_b t + q_c}$ in the segment's own parameter with $|\vec{e}|$ carried outside. Differentiating *that* form in (q_a, q_b, q_c) gives expressions with $q_a^{13/2}$ in the denominator. Change to arc length first. Let $\hat{u} = \vec{e}/|\vec{e}|$, $\vec{\delta} = \vec{a}^+ - \vec{z}_B$, $s = \vec{\delta} \cdot \hat{u}$, $\vec{\delta}_\perp = \vec{\delta} - s\hat{u}$, $h = |\vec{\delta}_\perp|$. Then $w = |\vec{e}| t + s$ absorbs the speed into the measure and

$$|\vec{e}| \int_{t_0}^{t_1} (r - \sqrt{q_a t^2 + q_b t + q_c})^3 dt = G(r, h, w_0, w_1), \quad G = \int_{w_0}^{w_1} (r - R)^3 dw, \quad R = \sqrt{w^2 + h^2}. \quad (5)$$

Only five antiderivatives are needed, and they are the same five the *value* needs:

$$\int \frac{dw}{R} = \text{asinh} \frac{w}{h}, \quad \int dw = w, \quad \int R dw, \quad \int R^2 dw = \frac{w^3}{3} + h^2 w, \quad \int R^3 dw. \quad (6)$$

Writing ΔF for $F(w_1) - F(w_0)$,

$$\frac{\partial G}{\partial r} = 3r^2 \Delta w - 6r \Delta \int R + 3 \Delta \int R^2, \quad (7)$$

$$\frac{\partial G}{\partial h} = -3h \left[r^2 \Delta \text{asinh} \frac{w}{h} - 2r \Delta w + \Delta \int R \right], \quad (8)$$

$$\frac{\partial G}{\partial w_1} = (r - R(w_1))^3, \quad \frac{\partial G}{\partial w_0} = -(r - R(w_0))^3, \quad (9)$$

the last two by the fundamental theorem, so they cost nothing. The middle line follows from $\partial R / \partial h = h/R$ and $(r - R)^2 / R = r^2 / R - 2r + R$. $h \rightarrow 0$ (arc center on the chord's line) is removable: $h \text{asinh}(w/h) \rightarrow 0$, so $\partial G / \partial h \rightarrow 0$.

Chaining back, with $\vec{\delta} = \vec{a}^+ - \vec{z}_B$ and $\vec{e} = \vec{a}^- - \vec{a}^+$,

$$\frac{\partial G}{\partial \vec{\delta}} = W \hat{u} + \frac{\partial G}{\partial h} \frac{\vec{\delta}_\perp}{h}, \quad W = \frac{\partial G}{\partial w_0} + \frac{\partial G}{\partial w_1}, \quad (10)$$

$$\frac{\partial G}{\partial \vec{e}} = S \hat{u} + \frac{1}{|\vec{e}|} \left(W \vec{\delta}_\perp - \frac{s}{h} \frac{\partial G}{\partial h} \vec{\delta}_\perp \right), \quad S = t_0 \frac{\partial G}{\partial w_0} + t_1 \frac{\partial G}{\partial w_1}, \quad (11)$$

using $\partial s / \partial \vec{e} = \vec{\delta}_\perp / |\vec{e}|$ and $\partial h / \partial \vec{e} = -s \vec{\delta}_\perp / (h |\vec{e}|)$.

(3) Arc along, segment to. The integrand is $(a + b \cos \psi + c \sin \psi)^3$ with $a = \vec{n} \cdot (\mathbf{tail}_B - \vec{z})$, $b = -\vec{n} \cdot \vec{w}_0$, $c = -\vec{n} \cdot J\vec{w}_0$, and the contribution is $\rho \operatorname{sgn}(\psi_k) [F(\psi_k t_1) - F(\psi_k t_0)]$. With $F = \int (a + b \cos \psi + c \sin \psi)^3 d\psi$,

$$F_a = 3a^2 \psi + 6a(b \sin \psi - c \cos \psi) + \frac{3}{2} \psi (b^2 + c^2) + \frac{3}{4} (b^2 - c^2) \sin 2\psi - \frac{3}{2} bc \cos 2\psi, \quad (12)$$

$$F_b = 3a^2 \sin \psi + \frac{3}{2} a(2b\psi + b \sin 2\psi - c \cos 2\psi) + b^2 (\cos^2 \psi + 2) \sin \psi - 2bc \cos^3 \psi + c^2 \sin^3 \psi, \quad (13)$$

$$F_c = -3a^2 \cos \psi - \frac{3}{2} a(b \cos 2\psi - 2c\psi + c \sin 2\psi) - b^2 \cos^3 \psi + 2bc \sin^3 \psi + c^2 (\cos^2 \psi - 3) \cos \psi, \quad (14)$$

$$F_\psi = (a + b \cos \psi + c \sin \psi)^3. \quad (15)$$

F_a is $\int 3(\cdot)^2 d\psi$ term by term; F_ψ is the fundamental theorem and is the cheapest available check on the other three. The sweep derivative is $t_1 F_\psi(\psi_k t_1) - t_0 F_\psi(\psi_k t_0)$. [These are derivatives of the same antiderivative the value uses, so the additive constant cancels in the difference and never needs tracking.](#)

(4) Arc along, arc to. $\int \sqrt{A + B \cos \psi} d\psi$ is an incomplete elliptic integral, so this one is Gauss–Legendre. The integrand is analytic on a sub-stretch — that is exactly what the partition guarantees — so it is differentiated *pointwise* at the nodes and the same weights carry the derivative. The derivative is then exact for the same reason the value is.

4 The normal pullback

Cases (1) and (3) both differentiate through $\vec{n} = K\vec{e}/|\vec{e}|$. Since $d\vec{n} = K d\vec{e}/|\vec{e}| - \vec{n}(\vec{e} \cdot d\vec{e})/|\vec{e}|^2$, the covector conjugate to $d\vec{e}$ is

$$\frac{1}{|\vec{e}|} \left[K^\top \vec{g} - (\vec{g} \cdot \vec{n}) \frac{\vec{e}}{|\vec{e}|} \right], \quad K^\top \vec{g} = (-g_y, g_x), \quad (16)$$

where $\vec{g}$ collects the coefficients conjugate to $\vec{n}$ ($\vec{g} = P_\alpha(\mathbf{tail}_B - \vec{a}^+) + P_m \vec{e}$ in case (1), $\vec{g} = F_a(\mathbf{tail}_B - \vec{z}) - F_b \vec{w}_0 - F_c J\vec{w}_0$ in case (3)). [The two terms carry different powers of \$|\vec{e}|\$. Factoring a single \$1/|\vec{e}|\$ out of both is the obvious slip and is nearly invisible — measured, it left one component 1.6% wrong and the other completely wrong, so a spot check on the wrong component reads as a pass. It was the only bug in the whole derivation.](#)

5 Area: everything is elementary

For a segment piece, $\vec{v} = (1-t) d\vec{a}^+ + t d\vec{a}^-$ and $\vec{n} ds = K\vec{e} dt$, so

$$\int (\vec{v} \cdot \vec{n}) ds = K\vec{e} \cdot \left[(\Delta_1 - \frac{1}{2}\Delta_2) d\vec{a}^+ + \frac{1}{2}\Delta_2 d\vec{a}^- \right]. \quad (17)$$

For an arc piece, write $\vec{w}(t) = \cos(\psi t) \vec{w}_0 + \sin(\psi t) J\vec{w}_0$, so $\vec{x} = \vec{z} + \vec{w}$, $\vec{n} = \vec{w}/\rho$ and $ds = \rho|\psi| dt$. Two identities collapse it:

$$\vec{w} \cdot (-\sin(\psi t) \vec{w}_0 + \cos(\psi t) J\vec{w}_0) = 0, \quad (\cos(\psi t) I + \sin(\psi t) J)^\top \vec{w} = \operatorname{Rot}(-\psi t) \vec{w} = \vec{w}_0. \quad (18)$$

The first says the tangential velocity has no normal component, so $\partial/\partial\psi$ vanishes *identically* rather than merely being small; the second kills all the trigonometry in the $\vec{w}_0$ term. What is left is

$$\int (\vec{v} \cdot \vec{n}) ds = |\psi| \left[d\vec{z} \cdot \int_{t_0}^{t_1} \vec{w} dt + \Delta_1 d\vec{w}_0 \cdot \vec{w}_0 \right], \quad \int \vec{w} dt = \frac{\sin \psi t_1 - \sin \psi t_0}{\psi} \vec{w}_0 + \frac{\cos \psi t_0 - \cos \psi t_1}{\psi} J\vec{w}_0. \quad (19)$$

Note also that $\vec{x}$ contains neither ρ nor ψ , so $\partial|A \cap B|/\partial\rho_k = 0$ and $\partial|A \cap B|/\partial\psi_k = 0$ at the body level — all of the ρ dependence reaches the area through $\vec{z}$ and the kiss points. So the area gradient needs no quadrature at all, and is therefore *exacter* than the order-12 Gauss–Legendre it replaces, as well as cheaper: the arc integrand is trigonometric, and that quadrature was very accurate on it but never exact.

6 From body arrays back to the backbone

The corner map $(\vec{v}_{k-1}, \vec{v}_k, \vec{v}_{k+1}, \rho_k) \mapsto (\vec{z}_k, \rho_k, \psi_k, \vec{a}_k^\pm)$ is local and cheap, so its Jacobian is still taken by complex step — but *once per body* per force evaluation, $3n$ evaluations of a vectorized closed form, not once per pair. Every pair scatters into $\partial E/\partial(\text{body})$ and each body is converted at the end. That ordering is what makes the corner map cost $O(\text{bodies})$ rather than $O(\text{pairs})$, and it is the ordering a kernel wants.

7 Status

Verified in `tests/roundedContactGradientCheck.py`, innermost first: each coefficient partial against a complex step of its own value routine (10^{-15}); the body-array gradient against a complex step of the same energy written in the body arrays (10^{-15} to 10^{-14}); the assembled gradient against the old per-degree-of-freedom complex step (10^{-16} to 10^{-13}), with both agreeing with a central difference of the *true* energy at the difference’s own floor; and both packing drivers against the assemblies they replace (10^{-15}). At $\rho = 0$ the area tier still reproduces `energies.sharpOverlapEnergyForce` to 3×10^{-15} in force.

At $N = 11$ the depth tier went 1.39s to 0.27s per force evaluation and the area tier 0.39s to 0.14s. The gradient is no longer what costs: `substretches` is 82% of the depth driver, over half of that inside `numpy.roots`. The switch equations are quartics after $u = \tan(p/2)$, so the next worthwhile step is a closed-form quartic in `_solveTrig` — which is also what the device needs, since a companion-matrix eigensolve is not something a kernel should be doing.